Discrete Probability Problems

Nine problems from ICPC-style contests, with solutions

These problems come from programming contests. Each statement keeps the probability core of the original and drops the programming layer (large inputs, modular output, data structures). Only discrete probability is used. Problems 1–5 are the main set, in course order; Problems 6–9 are backups. The selection is explained in `prob_course_candidates.md`. Every numerical answer below was checked by exact enumeration, and each model was checked against the original problem's official sample outputs.

Notation: $H_m = \sum_{i=1}^{m} \frac{1}{i}$ and $H_m^{(2)} = \sum_{i=1}^{m} \frac{1}{i^2}$; $[z^N]\, f(z)$ is the coefficient of z^N in $f(z)$.

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Main problems

1 Mingle

Source: NAQ 2025, Problem G (QOJ). *Topics:* indicator variables, linearity of expectation, Poisson limit.

Problem. Let $k \geq 1$ and $n \geq 2k+1$. There are n players and n rooms, both labelled by $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$. The rooms stand in a circle, so room $n-1$ is next to room 0. Player i walks into a room chosen uniformly at random from the $2k+1$ rooms $i-k, \ldots, i+k$ (indices mod n), independently of the other players. A player who ends up alone in a room is a *winner*. Let W be the number of winners.

(a) Find $\mathbb{E}[W]$. Evaluate it for $n = 3$, $k = 1$.

(b) Find $\lim_{k\to\infty} \mathbb{E}[W]/n$ and explain the limit with the Poisson approximation to the binomial distribution.

Solution. (a) Count rooms, not players. A winner is alone in a room, and a room with exactly one player holds exactly one winner. Hence

$$W = \sum_{r \in \mathbb{Z}_n} X_r, \qquad X_r = \mathbf{1}[\text{room } r \text{ holds exactly one player}].$$

Fix a room r. Player i can enter room r only if $r \in \{i-k, \ldots, i+k\}$, that is, $i \in \{r-k, \ldots, r+k\}$. These are $2k+1$ distinct players because $n \geq 2k+1$. Each of them enters room r with probability $\frac{1}{2k+1}$, independently of the others. So the number of players in room r is $N_r \sim \mathrm{Bin}\big(2k+1, \frac{1}{2k+1}\big)$, and

$$\mathbb{P}(X_r = 1) = \mathbb{P}(N_r = 1) = (2k+1) \cdot \frac{1}{2k+1}\Big(1 - \frac{1}{2k+1}\Big)^{2k} = \Big(\frac{2k}{2k+1}\Big)^{2k}.$$

The indicators X_r are dependent, but linearity of expectation does not need independence:

$$\mathbb{E}[W] = n\Big(\frac{2k}{2k+1}\Big)^{2k}.$$

For $n = 3$, $k = 1$ this is $3 \cdot (2/3)^2 = 4/3$.

(b) Put $M = 2k+1$. Then $\mathbb{E}[W]/n = (1 - 1/M)^{M-1} \to e^{-1}$ as $k \to \infty$. The reason is that $N_r \sim \mathrm{Bin}(M, 1/M)$ converges in distribution to a Poisson(1) variable, and $\mathbb{P}(\mathrm{Poisson}(1) = 1) = e^{-1}$. So in a large game about 37% of the rooms hold a winner. □

Remark. The answer does not depend on n. If every player instead chooses among all n rooms, the same argument gives the classical value $n(1-1/n)^{n-1}$.

2 Accidental Arithmetic

Source: NWERC 2025, Problem A (QOJ). *Topics:* linearity of expectation, symmetry, conditioning.

Problem. A positive integer N has decimal digits $d_1 d_2 \cdots d_L$ ($d_1 \neq 0$). You type the digits from left to right into a faulty calculator. Into each of the $L-1$ gaps between consecutive digits it inserts, independently, a "+" with probability β, a "−" with probability γ, and nothing with probability $q = 1 - \beta - \gamma$, where $\beta + \gamma > 0$. The resulting expression, such as $12 - 3 + 45$, is then evaluated; a term may begin with zeros, so $1 + 05 = 6$. Let V be the value, and let P_l be the number formed by the first l digits (so $P_L = N$).

(a) Let $\beta = \gamma = \alpha$. Show that

$$\mathbb{E}[V] = \sum_{l=1}^{L-1} 2\alpha(1-2\alpha)^{l-1} P_l + (1-2\alpha)^{L-1} N,$$

and evaluate it for $N = 12345$, $\alpha = 0.45$.

(b) Now let β, γ be arbitrary. Let F be the first term of the expression and S the sum of all its terms, each taken with a plus sign. Show that $\mathbb{E}[V] = (1-\mu)\mathbb{E}[F] + \mu\,\mathbb{E}[S]$ with $\mu = \frac{\beta-\gamma}{\beta+\gamma}$, and compute $\mathbb{E}[F]$ and $\mathbb{E}[S]$.

Solution. Number the gaps $1, \dots, L-1$, gap g lying between d_g and d_{g+1}. Call a gap a *sign gap* if it receives a sign, and let A be the (random) set of sign gaps. The set A cuts the digits into terms $F = T_0, T_1, \dots, T_s$, and

$$V = F + \sum_{t=1}^{s} \sigma_t T_t,$$

where $\sigma_t \in \{+1, -1\}$ is the sign in front of T_t. Given A, the terms T_t are fixed numbers, and the signs at the gaps in A are independent, each equal to $+1$ with probability $\beta/(\beta+\gamma)$. So $\mathbb{E}[\sigma_t \mid A] = \mu$ and

$$\mathbb{E}[V \mid A] = F + \mu \sum_{t=1}^{s} T_t = (1-\mu)F + \mu S.$$

Taking expectations gives $\mathbb{E}[V] = (1-\mu)\mathbb{E}[F] + \mu\mathbb{E}[S]$, which is the identity in (b).

The first term. Let G be the first sign gap, with $G = L$ if there is none. Then $F = P_G$, $\mathbb{P}(G = l) = (1-q)q^{l-1}$ for $l < L$, and $\mathbb{P}(G = L) = q^{L-1}$. Hence

$$\mathbb{E}[F] = \sum_{l=1}^{L-1} (1-q) q^{l-1} P_l + q^{L-1} N.$$

(a) If $\beta = \gamma = \alpha$ then $\mu = 0$: every later term is equally likely to be added or subtracted, whatever its digits, so only the first term survives. With $1 - q = 2\alpha$ the formula above is the claimed one. For $N = 12345$ and $\alpha = 0.45$ we have $q = 0.1$, and

$$\begin{aligned}\mathbb{E}[V] &= 0.9 \cdot 1 + 0.09 \cdot 12 + 0.009 \cdot 123 + 0.0009 \cdot 1234 + 0.0001 \cdot 12345 \\ &= 0.9 + 1.08 + 1.107 + 1.1106 + 1.2345 = 5.4321.\end{aligned}$$

(b) It remains to find $\mathbb{E}[S]$; use linearity over digits. Digit d_l contributes $d_l \cdot 10^{J_l}$ to S, where J_l is the number of digits after position l in the same term. That is, J_l counts the gaps $l, l+1, \dots$ that are empty before the first sign gap, capped at $L - l$. So $\mathbb{P}(J_l = j) = (1-q)q^j$ for $0 \le j < L - l$ and $\mathbb{P}(J_l = L - l) = q^{L-l}$, and

$$\mathbb{E}[S] = \sum_{l=1}^{L} d_l \Big(\sum_{j=0}^{L-l-1} (1-q)(10q)^j + (10q)^{L-l} \Big).$$

Check: if $q = 0$, every gap holds a sign, so $\mathbb{E}[F] = d_1$, $\mathbb{E}[S] = \sum_l d_l$ and $\mathbb{E}[V] = d_1 + \mu \sum_{l \ge 2} d_l$, as it should be. □

3 502 Bad Gateway

Source: ICPC Asia EC Online 2024 (II), Problem L (QOJ). *Topics:* geometric distribution, optimal stopping.

Problem. Fix an integer $T \geq 1$. A timer is set to a value drawn uniformly from $\{1, \ldots, T\}$ and counts down by 1 each second; you are done when it reaches 0. At the end of every second at which the timer is still positive, you may *reset* it: it is then set to a fresh uniform value from $\{1, \ldots, T\}$, independent of the past. You always see the timer. You want to minimise the expected time until you are done.

(a) For $1 \leq c \leq T$, consider the policy "keep a drawn value if it is at most c; otherwise wait one second and reset". Show that its expected time is

$$g(c) = \frac{T}{c} + \frac{c-1}{2}.$$

(b) Show that some policy of this form is optimal.

(c) Find the optimal c and the minimal expected time. Evaluate it for $T = 3$, 10 and 1000, and find its asymptotics as $T \to \infty$.

Solution. (a) Each draw is kept with probability c/T, independently of the other draws, so the number of rejected draws is geometric with mean $\frac{1-c/T}{c/T} = \frac{T-c}{c}$. A rejected draw is larger than $c \geq 1$, so the timer is still positive after one second, and the draw costs exactly one second. The kept draw is uniform on $\{1, \ldots, c\}$ and costs its value, with mean $\frac{c+1}{2}$. Hence

$$g(c) = \frac{T-c}{c} + \frac{c+1}{2} = \frac{T}{c} + \frac{c-1}{2}.$$

(b) Let V be the minimal expected time measured from the moment a fresh value is drawn; $V \leq g(T) = \frac{T+1}{2}$ is finite. After a reset the future is a fresh copy of the whole problem, so the best you can do from a reset is V, whatever happened before.

Suppose that at the end of some second the timer shows $s \geq 1$. If you never reset again, you need s more seconds. If you reset after waiting $w \geq 0$ more seconds, you need at least $w + V \geq V$, with equality for $w = 0$. So the best you can do from this point is $\min(s, V)$: reset now if $s > V$, and never reset if $s \leq V$.

After a draw $u \geq 2$, the first decision point shows $s = u - 1$. So it is optimal to keep u exactly when $u \leq V + 1$. Once a value is kept, later decision points show smaller values, which are still at most V, so it stays kept. (A draw $u = 1$ ends the game at once.) This is the threshold policy with $c = \min\bigl(T, \lfloor V + 1 \rfloor\bigr)$. Therefore $V = \min_{1 \leq c \leq T} g(c)$.

(c) For $1 \leq c < T$,

$$g(c+1) - g(c) = \frac{1}{2} - \frac{T}{c(c+1)}.$$

So g decreases while $c(c+1) < 2T$ and does not decrease afterwards. The optimal threshold is the least c with $c(c+1) \geq 2T$, namely $c^* = \bigl\lceil (\sqrt{8T+1} - 1)/2 \bigr\rceil$; when $c^*(c^*+1) = 2T$ the values c^* and $c^* + 1$ tie.

- $T = 3$: $c^* = 2$ (tied with 3), $V = \frac{3}{2} + \frac{1}{2} = 2$.
- $T = 10$: $c^* = 4$ (tied with 5), $V = \frac{10}{4} + \frac{3}{2} = 4$.
- $T = 1000$: $44 \cdot 45 = 1980 < 2000 \leq 45 \cdot 46$, so $c^* = 45$ and $V = \frac{1000}{45} + 22 = \frac{398}{9} \approx 44.22$.

For large T, the function $x \mapsto T/x + x/2$ is minimised at $x = \sqrt{2T}$ with value $\sqrt{2T}$, and rounding x to an integer changes the value by $O(T^{-1/2})$. Hence $c^* = \sqrt{2T} + O(1)$ and $V = \sqrt{2T} - \frac{1}{2} + O(T^{-1/2})$; for $T = 1000$ this gives 44.22. Never resetting would cost $\frac{T+1}{2}$, so resets are worth much more than the second each one wastes. □

4 Uniform Chemistry

Source: SWERC 2022–2023, Problem B (QOJ). *Topics:* Markov chains, independent indicators, random permutations.

Problem. Fix integers $1 \le a < n$ and put $m = n - a$. A particle starts at a. At each step, if it is at $x < n$, it jumps to a point chosen uniformly from $\{x+1, \dots, n\}$, independently of the past. Let T be the number of steps until it reaches n.

(a) For $a < j < n$, let I_j be the indicator that the particle visits j. Show that $\mathbb{P}(I_j = 1) = \frac{1}{n-j+1}$ and that $I_{a+1}, \dots, I_{n-1}$ are independent.

(b) Find $\mathbb{E}[T]$ and $\operatorname{Var}(T)$.

(c) Let $c(m,t)$ be the unsigned Stirling numbers of the first kind, defined by $\sum_t c(m,t) z^t = z(z+1)\cdots(z+m-1)$; $c(m,t)$ is the number of permutations of m elements with exactly t cycles. Show that $\mathbb{P}(T = t) = c(m,t)/m!$.

(d) Three researchers each run an independent copy of this process with $a = 1$ and $n = 3$, one step per year. A researcher wins if nobody reaches n strictly earlier, so ties all win. Find the probability that a given researcher wins.

Solution. (a) Compute the joint law directly. Let $S = \{s_1 < \cdots < s_r\} \subseteq \{a+1, \dots, n-1\}$. The particle visits exactly the points of S between a and n if and only if its path is $a \to s_1 \to \cdots \to s_r \to n$, which has probability

$$\frac{1}{n-a} \prod_{i=1}^{r} \frac{1}{n-s_i}.$$

Now put $q_j = \frac{1}{n-j+1}$. The product $\prod_{j=a+1}^{n-1}(1-q_j) = \prod_{j=a+1}^{n-1} \frac{n-j}{n-j+1}$ telescopes to $\frac{1}{n-a}$, and $\frac{q_j}{1-q_j} = \frac{1}{n-j}$. Hence

$$\prod_{j \in S} q_j \prod_{j \notin S} (1-q_j) = \prod_{j=a+1}^{n-1} (1-q_j) \cdot \prod_{j \in S} \frac{q_j}{1-q_j} = \frac{1}{n-a} \prod_{j \in S} \frac{1}{n-j},$$

which is the same probability. So $(I_{a+1}, \dots, I_{n-1})$ has the law of independent Bernoulli(q_j) variables, and both claims follow.

There is also a direct argument for the marginal: the particle must enter $\{j, \dots, n\}$ at some step, and the jump that first enters this set lands uniformly on its $n - j + 1$ points.

(b) Each step lands on a new point and the last step lands on n, so $T = 1 + \sum_{j=a+1}^{n-1} I_j$. Substitute $i = n - j + 1$, which runs over $2, \dots, m$:

$$T = 1 + \sum_{i=2}^{m} B_i, \qquad B_i \sim \text{Bernoulli}(1/i) \text{ independent.}$$

Hence $\mathbb{E}[T] = 1 + \sum_{i=2}^{m} \frac{1}{i} = H_m$ and, by independence,

$$\operatorname{Var}(T) = \sum_{i=2}^{m} \frac{1}{i}\Big(1 - \frac{1}{i}\Big) = H_m - H_m^{(2)}$$

(the term $i = 1$ of $H_m - H_m^{(2)}$ is zero).

(c) By independence,

$$\mathbb{E}\big[z^T\big] = z \prod_{i=2}^{m} \Big(1 - \frac{1}{i} + \frac{z}{i}\Big) = z \prod_{i=2}^{m} \frac{z+i-1}{i} = \frac{z(z+1)\cdots(z+m-1)}{m!} = \sum_t \frac{c(m,t)}{m!} z^t.$$

Comparing coefficients gives $\mathbb{P}(T=t) = c(m,t)/m!$. So T has the law of the number of cycles of a uniformly random permutation of m elements.

(d) With $a=1$ and $n=3$, the first jump goes to 3 with probability $\frac{1}{2}$ (then $T=1$); otherwise it goes to 2 and then to 3 (then $T=2$). A given researcher wins if their $T=1$, or if their $T=2$ and both other researchers also have $T=2$:

$$\mathbb{P}(\text{win}) = \frac{1}{2} + \left(\frac{1}{2}\right)^3 = \frac{5}{8}.$$

In general, researcher i wins with probability $\sum_t \mathbb{P}(T_i=t)\prod_{j\neq i}\mathbb{P}(T_j \geq t)$, and (c) gives each distribution. □

5 Jinx or Jackpot

Source: NERC 2025, Problem J (QOJ). *Topics:* Bayes' rule, dynamic programming, value of information.

Problem. A slot machine pays even odds: if you bet x, you win x with probability p (a *jackpot*) and lose x with probability $1-p$ (a *jinx*). Given p, the rounds are independent. You start with \$1000 and may play at most k rounds. In each round you choose a bet x with $0 \leq x \leq$ (current balance); a bet of 0 is allowed, and you still see whether that round was a jackpot. You want to maximise the expected final balance.

(a) Show that the optimal expected final balance from balance w is w times a number that depends only on the outcomes seen so far, and that in every round an optimal bet is either 0 or the whole balance.

(b) Suppose p is known. Find an optimal strategy and the maximal expected profit. Evaluate the profit for $p=0.61$, $k=5$.

(c) Suppose p is unknown: it is 0.7 or 0.3, each with prior probability $\frac{1}{2}$. Let $k=2$. Find an optimal strategy and the maximal expected profit, and compare it with the best strategy that bets 0 in the first round.

Solution. (a) Let h be the sequence of outcomes seen so far, and let j be the number of rounds left. Given h, the next round is a jackpot with probability $\hat{p}(h) = \mathbb{E}[p \mid h]$, the posterior mean of p (law of total probability). Write hW and hL for h followed by a jackpot or a jinx. Let $V_j(w,h)$ be the optimal expected final balance. We show by induction on j that $V_j(w,h) = w\,v_j(h)$, where $v_0 \equiv 1$ and

$$v_j(h) = \max\Big\{\hat{p}\,v_{j-1}(hW) + (1-\hat{p})\,v_{j-1}(hL),\ 2\hat{p}\,v_{j-1}(hW)\Big\}, \qquad \hat{p} = \hat{p}(h).$$

For $j=0$ this is clear. For $j \geq 1$, a bet $x \in [0,w]$ is followed by balance $w+x$ after a jackpot and $w-x$ after a jinx, and the outcome is seen either way, so by induction

$$V_j(w,h) = \max_{0\leq x\leq w}\Big[\hat{p}\,(w+x)\,v_{j-1}(hW) + (1-\hat{p})\,(w-x)\,v_{j-1}(hL)\Big].$$

The bracket is linear in x, so its maximum is attained at $x=0$ or $x=w$. These two choices give the two expressions in the formula for v_j.

(b) If p is known, then $\hat{p}=p$ in every round and $v_j = v_{j-1}\max(1,2p)$, so $v_k = \max(1,2p)^k$. If $p > \frac{1}{2}$, go all-in every round; the expected profit is $1000\big((2p)^k - 1\big)$. If $p \leq \frac{1}{2}$, never bet; the profit is 0. For $p=0.61$ and $k=5$ the profit is $1000\,(1.22^5 - 1) = 1000 \cdot 1.7027081632 \approx \$1702.71$.

(c) Bayes' rule after one jackpot:

$$\mathbb{P}(p=0.7 \mid W) = \frac{\frac{1}{2}\cdot 0.7}{\frac{1}{2}\cdot 0.7 + \frac{1}{2}\cdot 0.3} = 0.7, \qquad \hat{p}(W) = 0.7\cdot 0.7 + 0.3\cdot 0.3 = 0.58.$$

Likewise $\mathbb{P}(p = 0.7 \mid L) = 0.3$ and $\hat{p}(L) = 0.3 \cdot 0.7 + 0.7 \cdot 0.3 = 0.42$. Before any round, $\hat{p} = 0.5$.

Round 2. After a jackpot, $v_1(W) = \max(1,\, 2 \cdot 0.58) = 1.16$: go all-in. After a jinx, $v_1(L) = \max(1,\, 2 \cdot 0.42) = 1$: bet nothing.

Round 1. Betting 0 gives $\frac{1}{2} \cdot 1.16 + \frac{1}{2} \cdot 1 = 1.08$. Going all-in gives $2 \cdot \frac{1}{2} \cdot 1.16 = 1.16$. So the optimal strategy is to go all-in in round 1 and, after a jackpot, all-in again. The maximal expected profit is $1000\,(1.16 - 1) = \$160$. The best strategy that only watches round 1 earns $1000\,(1.08 - 1) = \$80$.

Round 1 is a fair bet ($\hat{p} = \frac{1}{2}$), yet staking everything on it is optimal. A jackpot both doubles the balance and makes the machine look better, so the balance is largest exactly when the next bet is most favourable. □

Backup problems

6 Orchard Fence

Source: 2025 ICPC Southeast USA Regional, Division 1, Problem A; listed under Mid-Atlantic USA in `data/tagged.json` (QOJ). *Topics:* linearity of expectation over edges, counting.

Problem. Let $z_1, \dots, z_n$ be $n \geq 3$ points in the plane in general position (no three on a line). A uniformly random subset K of k of them is kept, where $2 \leq k \leq n$. Let L be the perimeter of the convex hull of K; for $k = 2$ the hull is a segment, and its perimeter is twice its length.

(a) For an ordered pair (i, j) with $i \neq j$, let ℓ_{ij} be the number of the other $n - 2$ points that lie strictly to the left of the directed line from z_i to z_j. Show that

$$\mathbb{E}[L] = \frac{1}{\binom{n}{k}} \sum_{i \neq j} |z_i z_j| \binom{\ell_{ij}}{k-2}.$$

(b) Evaluate $\mathbb{E}[L]$ for the six vertices of a regular hexagon with side 1, for $k = 3$ and for $k = 4$.

Solution. (a) Traverse the hull boundary counterclockwise. Then

$$L = \sum_{i \neq j} |z_i z_j| \cdot \mathbf{1}[z_i \to z_j \text{ is a counterclockwise hull edge of } K]\,.$$

For $k \geq 3$ in general position, $z_i \to z_j$ is a counterclockwise hull edge exactly when $i, j \in K$ and every other point of K lies strictly to the left of the line $z_i \to z_j$. The number of such k-subsets is $\binom{\ell_{ij}}{k-2}$: choose the other $k - 2$ kept points among the ℓ_{ij} points on the left. For $k = 2$ the formula counts the segment in both directions, since $\binom{\ell}{0} = 1$, which matches the convention. Linearity of expectation gives the formula.

(b) Label the vertices $0, \dots, 5$ counterclockwise. There are three kinds of pairs:

- 6 adjacent pairs, length 1: $\ell = 4$ in the counterclockwise direction and $\ell = 0$ in the other;
- 6 pairs one vertex apart, length $\sqrt{3}$: $\ell = 3$ and $\ell = 1$;
- 3 opposite pairs, length 2: $\ell = 2$ in both directions.

For $k = 3$, the weights $\binom{\ell}{1}$ of the two directions add up to 4 for every pair, so

$$\mathbb{E}[L] = \frac{4\,(6 \cdot 1 + 6\sqrt{3} + 3 \cdot 2)}{\binom{6}{3}} = \frac{48 + 24\sqrt{3}}{20} = \frac{12 + 6\sqrt{3}}{5} \approx 4.478.$$

Check: a triangle's perimeter is the sum of its three sides, so $\mathbb{E}[L]$ is three times the mean distance of a random pair, $3 \cdot (12 + 6\sqrt{3})/15$.

For $k = 4$, the weights $\binom{\ell}{2}$ add up to $6 + 0$, $3 + 0$ and $1 + 1$ for the three kinds, so

$$\mathbb{E}[L] = \frac{6 \cdot 1 \cdot 6 + 6\sqrt{3} \cdot 3 + 3 \cdot 2 \cdot 2}{\binom{6}{4}} = \frac{48 + 18\sqrt{3}}{15} = \frac{16 + 6\sqrt{3}}{5} \approx 5.278.$$

□

Remark. In the original problem the kept points are tree trunks, disks of diameter d. The shortest fence around them has length $L + \pi d$, so the expected fence length is $\mathbb{E}[L] + \pi d$.

7 Chopsticks

Source: 2025 ICPC Asia Taiwan Online Programming Contest, Problem H (QOJ). *Topics:* parity, indicators, generating functions.

Problem. A box holds chopsticks of m types, k_i of type i, and $s = \sum_i k_i$ in total. You draw $2n \le s$ of them uniformly at random without replacement and hand them out in n pairs so that as many pairs as possible are of one type. Let M be the number of mixed pairs, and let X_i be the number of drawn chopsticks of type i.

(a) Show that $M = \frac{1}{2}\,\#\{i : X_i \text{ odd}\}$, and hence $\mathbb{E}[M] = \frac{1}{2}\sum_i \mathbb{P}(X_i \text{ odd})$.

(b) Show that $\mathbb{P}(X_i \text{ odd}) = \frac{1}{2}\big(1 - \mathbb{E}[(-1)^{X_i}]\big)$ and

$$\mathbb{E}\big[(-1)^{X_i}\big] = \frac{[z^{2n}]\,(1-z)^{k_i}(1+z)^{s-k_i}}{\binom{s}{2n}}.$$

(c) Find $\mathbb{E}[M]$ for 8 red and 8 blue chopsticks and $n = 5$.

(d) Suppose instead that the $2n$ chopsticks are drawn independently, each of type i with probability p_i. Find $\mathbb{E}[M]$.

Solution. (a) Type i gives at most $\lfloor X_i/2 \rfloor$ same-type pairs, and pairing within types achieves $\sum_i \lfloor X_i/2 \rfloor$. The leftover chopsticks, one of each type with X_i odd, are then paired arbitrarily. If O is the number of odd X_i, the matched pairs number $\sum_i \lfloor X_i/2 \rfloor = (2n - O)/2$, so $M = n - (2n - O)/2 = O/2$. Linearity gives $\mathbb{E}[M] = \frac{1}{2}\sum_i \mathbb{P}(X_i \text{ odd})$.

(b) Since $(-1)^X = 1 - 2 \cdot \mathbf{1}[X \text{ odd}]$, we get $\mathbb{P}(X \text{ odd}) = \frac{1}{2}\big(1 - \mathbb{E}[(-1)^X]\big)$. The count X_i is hypergeometric: $\mathbb{P}(X_i = x) = \binom{k_i}{x}\binom{s-k_i}{2n-x}\big/\binom{s}{2n}$. So

$$\binom{s}{2n}\mathbb{E}\big[(-1)^{X_i}\big] = \sum_x (-1)^x \binom{k_i}{x}\binom{s-k_i}{2n-x} = [z^{2n}]\,(1-z)^{k_i}(1+z)^{s-k_i},$$

because multiplying the two generating functions $(1-z)^{k_i} = \sum_x (-1)^x \binom{k_i}{x} z^x$ and $(1+z)^{s-k_i} = \sum_y \binom{s-k_i}{y} z^y$ collects exactly these products in the coefficient of z^{2n}.

(c) Here $k = 8$, $s = 16$ and $2n = 10$. Since $(1-z)^8(1+z)^8 = (1-z^2)^8$,

$$[z^{10}]\,(1-z^2)^8 = \binom{8}{5}(-1)^5 = -56, \qquad \mathbb{E}\big[(-1)^{X_{\text{red}}}\big] = \frac{-56}{\binom{16}{10}} = \frac{-56}{8008} = -\frac{1}{143}.$$

So $\mathbb{P}(X_{\text{red}} \text{ odd}) = \frac{1}{2}\big(1 + \frac{1}{143}\big) = \frac{72}{143}$, and the same holds for blue. Hence $\mathbb{E}[M] = \frac{1}{2} \cdot 2 \cdot \frac{72}{143} = \frac{72}{143} \approx 0.503$. (With two colours, M is simply the indicator that X_{red} is odd.)

(d) Now $X_i \sim \text{Bin}(2n, p_i)$, so $\mathbb{E}[(-1)^{X_i}] = \sum_x \binom{2n}{x}(-p_i)^x(1-p_i)^{2n-x} = (1-2p_i)^{2n}$, and

$$\mathbb{E}[M] = \frac{1}{4}\sum_i \Big(1 - (1-2p_i)^{2n}\Big).$$

□

8 Crashing Competition Computer

Source: NWERC 2022, Problem C (QOJ). *Topics:* consecutive successes, recursions for expectations, convexity.

Problem. You must type a text of c characters, one time unit per character. After each typed character, independently, the computer crashes with probability $p \in (0,1)$. A crash costs r time units of recovery, after which you continue from the last save; the start of the text counts as saved. A save takes t time units and never crashes, and the text must be saved after its last character. Put $q = 1 - p$.

(a) Show that typing a block of k characters that starts right after a save and ends with a save takes expected time $h(k) + t$, where

$$h(k) = \Big(\frac{1}{p} + r\Big)\big(q^{-k} - 1\big).$$

(b) Show that, for a fixed number of saves, the expected total time is smallest when the block lengths differ by at most one.

(c) Find the minimal expected time for $c = 3$, $p = \frac{1}{2}$, $r = 2$, $t = 5$.

Solution. (a) Let E_j be the expected time, starting from the save, until j characters of the block have been typed with no crash in between. Once $k-1$ characters are typed, you type one more (one time unit). With probability p it crashes; you then pay r and are back at the save, so you need E_k more in expectation. Hence

$$E_k = E_{k-1} + 1 + p\,(r + E_k), \qquad \text{so} \qquad E_k = \frac{E_{k-1} + 1 + pr}{q}, \qquad E_0 = 0.$$

Unrolling the recursion,

$$E_k = (1 + pr)\sum_{i=1}^{k} q^{-i} = (1 + pr)\,\frac{q^{-k} - 1}{1 - q} = \Big(\frac{1}{p} + r\Big)\big(q^{-k} - 1\big) = h(k),$$

and the final save adds t.

(b) First, a fixed plan of save positions is enough. A crash returns you to the state you were in right after the last save, so where to save next from a given save position can be decided once and for all. The plan is a sequence $0 = y_0 < y_1 < \cdots < y_B = c$ of save positions, with block lengths $k_b = y_b - y_{b-1}$. By (a) and linearity its expected time is $\sum_{b=1}^{B} h(k_b) + Bt$.

The increments $\Delta(k) = h(k+1) - h(k) = \big(\frac{1}{p} + r\big)\frac{p}{q}\,q^{-k}$ increase with k, so h is convex. If two blocks have lengths $a \ge b + 2$, replacing them by $a - 1$ and $b + 1$ changes the total by

$$\Delta(b) - \Delta(a - 1) \le 0,$$

because $b < a - 1$. Repeating this until all lengths differ by at most one never increases the total. Writing $c = Bu + v$ with $0 \le v < B$, the best plan with B saves therefore costs $v\,h(u+1) + (B - v)\,h(u) + Bt$, and it remains to minimise over $B \in \{1, \dots, c\}$.

(c) Here $\frac{1}{p} + r = 4$ and $q^{-1} = 2$, so $h(k) = 4(2^k - 1)$: $h(1) = 4$, $h(2) = 12$, $h(3) = 28$.

- One save (block 3): $28 + 5 = 33$.
- Two saves (blocks $2, 1$): $12 + 4 + 2 \cdot 5 = 26$.
- Three saves (blocks $1, 1, 1$): $3 \cdot (4 + 5) = 27$.

The minimum is 26: save once after the second (or the first) character, and again at the end. □

9 Disbursement on Quarantine Policy

Source: ICPC Asia Taiwan 2024, Problem D (QOJ). *Topics:* linearity of expectation, complements, independence.

Problem. Passengers sit in a rectangular grid of seats, and each is infected independently with probability $\frac{1}{2}$. An infected passenger is quarantined for d_0 days. A healthy passenger with at least one infected edge-neighbour (sharing a side) is quarantined for d_1 days. A healthy passenger with no infected edge-neighbour but at least one infected corner-neighbour (sharing only a corner) is quarantined for d_2 days. Everyone else is not quarantined.

(a) A seat has e edge-neighbours and c corner-neighbours. Find the expected quarantine length of its passenger.

(b) Find the expected total number of quarantine days for a 3×3 grid with $d_0 = 4$, $d_1 = 2$, $d_2 = 1$.

Solution. (a) The events "the passenger is infected", "no edge-neighbour is infected" and "no corner-neighbour is infected" concern disjoint sets of seats, so they are independent, with probabilities $\frac{1}{2}$, 2^{-e} and 2^{-c}. Using complements,

$$\mathbb{P}(\text{healthy, some edge-neighbour infected}) = \tfrac{1}{2}\big(1 - 2^{-e}\big),$$
$$\mathbb{P}(\text{healthy, no edge-neighbour infected, some corner-neighbour infected}) = \tfrac{1}{2}\,2^{-e}\big(1 - 2^{-c}\big).$$

So the expected quarantine length is

$$\frac{d_0}{2} + \frac{d_1}{2}\big(1 - 2^{-e}\big) + \frac{d_2}{2}\,2^{-e}\big(1 - 2^{-c}\big).$$

(b) By linearity, add up the seats. With $d_0 = 4$, $d_1 = 2$, $d_2 = 1$:

- 4 corner seats ($e = 2$, $c = 1$): $2 + \frac{3}{4} + \frac{1}{16} = \frac{45}{16}$;
- 4 side seats ($e = 3$, $c = 2$): $2 + \frac{7}{8} + \frac{3}{64} = \frac{187}{64}$;
- 1 centre seat ($e = 4$, $c = 4$): $2 + \frac{15}{16} + \frac{15}{512} = \frac{1519}{512}$.

The expected total is

$$4 \cdot \frac{45}{16} + 4 \cdot \frac{187}{64} + \frac{1519}{512} = \frac{5760 + 5984 + 1519}{512} = \frac{13263}{512} \approx 25.90.$$

□

Remark. In the original problem some passengers are known to be infected or healthy. The same argument works with each seat's infection probability in $\{0, \frac{1}{2}, 1\}$; the probability that no edge-neighbour is infected becomes the product of $1 - \mathbb{P}(\text{infected})$ over the edge-neighbours.